Short-Form Video Platforms, Sustainability Content, and Purchase Intention Among Generation Z: A Cross-Platform Analysis of TikTok, Instagram Reels and YouTube Shorts

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Abstract*—The short-form video platforms have changed the way that Generation Z customers come into contact with sustainability content; however, the mechanism of going from sustainability content perception through algorithms to purchase intention has not yet been thoroughly researched. Although there are several pieces of research regarding the effects of social media on sustainable consumption, to the best of our knowledge, there is no piece of research studying the effects of TikTok, Instagram Reels, and YouTube Shorts in concert on Generation Z's sustainable purchase intention. The current paper investigates the effects of sustainability content exposure (SCE) through the stimulus-organism-response framework, with environmental awareness (EA), environmental attitude (AT), and perceived social norms (SN) acting as mediators. A quantitative cross-sectional survey was undertaken among 100 international students from four universities in Germany, and data were analysed using PLS-SEM with SmartPLS and 5,000 bootstrap subsamples. The results indicated that SCE had a positive relation to EA ($\beta = 0.901$), AT ($\beta = 0.854$), and SN ($\beta = 0.719$), all $p < 0.001$. Among the three mediating variables, only SN ($\beta = 0.763$, $p < 0.001$) and AT ($\beta = 0.391$, $p = 0.002$) positively influenced PI, while EA did not have any effect ($\beta = -0.201$, $p = 0.109$). The model explains 84.1% of the variance in PI and has predictive relevance (positive Q^2_{predict} for all endogenous indicators), and five out of six hypotheses are supported. Exploratory analysis reveals no difference in construct scores between the platforms. However, HTMT values are greater than 0.90 for four construct pairs, and collinearity is high between EA and AT (VIF > 5), which implies discriminant validity problems. The sample used is convenience (79% postgraduate and 51% Instagram users), which means poor generalisability. The study results indicate that social norms, rather than awareness, relate sustainability content to purchase intention.

***Keywords*—Generation Z, sustainability content, short-form video, purchase intention, TikTok, Instagram Reels, YouTube Shorts, S-O-R framework, PLS-SEM, SDG 12**

I. Introduction

Generation Z did not suddenly develop an interest in sustainability; they grew up with it, and evidence suggests that this shapes their purchasing decisions. Generation Z is more willing to spend extra on eco-friendly products as compared to the older generation [1]. Unlike the older generation, who may have picked up environmental awareness later in life, for Generation Z it has always been part of how they think about the brands and products [1]. Short-form video platforms work differently from older social media. The algorithm keeps pushing content to users based on what they already engage with; this means a young person can end up watching sustainability-related videos repeatedly without even searching for them, and research suggests that this kind of repeated exposure is what actually shifts attitudes over time. But how this daily exposure actually leads someone to go and buy something sustainable remains poorly understood. Research consistently shows that social media content shapes sustainable consumption behavior among Generation Z, but the exact process in between is not well explained. However, a systematic review of 92 studies found that existing research is heavily focused on brand-driven marketing strategies and largely ignores how non-brand sustainability content, including creator-driven short-form video, influences young consumers [2]. What is missing from existing research is a focus specifically on short-form video platforms. Most studies look at social media in general and confirm that it influences Generation Z's green purchasing behavior, but do not separate out platforms from everything else [1]–[3]. Some studies even mention cross-platform comparison as something future research should look into, which suggests researchers themselves recognize this gap [2]. This study addresses this gap by focusing specifically on short-form video exposure and examining how it relates to sustainable purchase intention among Generation Z through three internal pathways: environmental awareness, environmental attitude, and perceived social norms. This paper addresses the following research question: *How does exposure to sustainability-related content on short-form video platforms affect purchase intention among Generation Z consumers?*

The three platforms are analysed in one model, and platform differences are examined as an exploratory comparison, since the sample is too small for formal multi-group testing.

The following section reviews existing literature across four areas: Social media and consumer behavior, Generation Z and sustainable consumption, short-form video and purchase intention, and theoretical frameworks. From this review, the hypotheses are developed, followed by a description of the research methodology and survey design. The conceptual framework is subsequently presented, and the paper ends with a discussion of findings, limitations, and suggestions for future research. The current study contributes to the achievement of Sustainable Development Goals 12 (Responsible Consumption and Production) and 13 (Climate Action) of the United Nations by investigating the influence of digital content on the purchase intention of sustainable products and the way sustainability communication can motivate consumers' climate-related behavior.

II. Literature Review

Social media platforms have consistently shaped consumers' attitudes, whether they decide to buy products, and this holds across different product types or countries [4], [5]. One of the most commonly used frameworks to explain this is the Stimulus-Organism-Response (SOR) model. SOR is the dominant theoretical lens for this relationship, mapping the social media content as the stimulus, the person's internal thoughts and feelings as the middle step, and their purchasing behavior as the result [5], [6] . One study applied this framework to 660 Generation Z consumers in Italy and found that exposure to sustainability content on social media directly led to more sustainable habits and buying behavior, which supports the idea that the stimulus-response pathway works in an environmental context too [5]. Another study looked at how people adopt information from social media and found that the quality of the content, how useful it is, and how trustworthy the source seems all play a role in pushing people toward sustainable purchasing decisions [7]. So it is not just about seeing the content; what the content is actually like matters too. That said, most of these studies treat social media as one big category. They do not break it down by platform, and some researchers have pointed out that this is something future studies should look into [2].

A. Generation Z and Sustainable Consumption

Generation Z has been exposed to news and content regarding environmental issues throughout their lives, and studies have proved that they are more environmentally conscious compared to previous generations [1], [3]. Studies also show that attitude and intention significantly influence the relationship between social media exposure and sustainable actions. For instance, there are studies regarding Indonesian consumers where environmental concern and social media exposure positively predict sustainable consumption [8], [9].

This pattern is not limited to one country. Studies from China, Indonesia, and India all report similar findings. Young consumers who engage more with social media tend to show stronger intentions to buy sustainably [1], [3], [4]. However, the researchers do not fully agree on exactly how this works. Some studies find that green knowledge directly affects purchasing behavior among Generation Z [1], while others argue that the knowledge alone is not enough. Environmental attitude needs to be there as well, which acts as the bridge between knowing and actually doing [9].

There is also a well-known problem called the attitude-behavior gap. Even when Generation Z consumers hold positive environmental views, they do not always follow through with sustainable purchases. The reasons are high price, convenience, distrust of green claims, and various ways people justify not changing their habits [1], [10]. Some researchers suggest that social media and peer influence make a pivotal impact on making Generation Z's sustainable consumption choices [11].

B. Short Form Video Platforms and Purchase Intention

TikTok has been described as the 'TV for Generation Z' [12], and is considered one of the most important channels for reaching young consumers, with around 70% of Generation Z consumers reportedly using it [12]. A study conducted on 366 TikTok users in the US using the 'Theory of Planned Behavior' found that exposure to sustainability-themed TikTok videos significantly increased the intention to buy sustainable clothes [12]. A separate study of 241 Generation Z TikTok users in Germany found similar results, confirming that TikTok videos positively influence sustainable purchase intention [13]. Both studies only look at one platform. The US study is also limited to one product category and country, which makes it hard to generalize the findings [12], [13]. So while the evidence confirms that short-form video content does influence sustainable purchase intention, no study has yet compared TikTok, Instagram Reels, and YouTube Shorts together using Generation Z as the target group and sustainable content as the main variable. Recent studies show growing interest in this direction, but none yet closes the specific gap identified above. In one such 2025 study involving 643 Gen Z participants, both brand-created and user-created content on TikTok and Instagram were found to predict purchase intentions of sustainable fashion, with social media engagement serving as a mediator variable [14]. Another 2025 experimental study comparing green advertising on Instagram and TikTok found that message framing consistent with the genre of the platform led to enhanced processing fluency and purchase intentions, showing that platform-specific attributes, apart from mere exposure, influence the outcomes [15]. Moreover, studies on reception indicate that Gen Z is particularly sensitive to entertaining narratives used in environmental videos [16]. This gap is also identified by a systematic review of 92 studies [2].

C. Theoretical Framework

Three frameworks come up repeatedly in this area of research: Stimulus Organism Response (SOR), Theory of Planned Behavior (TPB), and the Knowledge Attitude Practice (KAP) framework [1], [5], [12]. SOR treats social media as a trigger that leads to internal processing and then behavioral outcomes like purchase intention. TPB focuses on the impact

of attitude, social norm, and perceived control on behavioral intentions [17]. KAP suggests that knowledge comes first, then attitude shift, and then behavior follows [5], [8]. This study uses SOR as the main framework because it fits best with what this research is trying to do: understand how sustainability content delivered through short video platforms ends up affecting purchase intention. Unlike TPB and KAP, SOR does not assume that users are making fully deliberate choices or that they need prior knowledge before the content affects them, which makes more sense for a Generation Z audience that is passively exposed to content across multiple platforms [5]. The hypotheses developed in this study are all built around this framework.

III. Hypotheses

Each of the six hypotheses maps to one arrow in the conceptual framework. Together, they cover the full SOR chain from sustainability content exposure through three internal states to purchase intention. Short-form video platforms push sustainability content to users repeatedly through the personalised feeds, which means users are more likely to actually process the information rather than just scroll past it. Research has shown that environmental content on social media significantly predicts sustainability intention among Generation Z (β =0.281; $p < 0.001$) [8], which suggests that simply being exposed to this content builds genuine awareness. Short-form video makes this effect even stronger because of how frequently the content is delivered through algorithms compared to the old static format [5].

> **H1:***Sustainability content exposure on short-form video platforms is positively associated with environmental awareness among Generation Z consumers.*

Social media does not seem to push people toward green purchasing directly. The evidence points to attitude as the step in between. The same attitudinal path has also been demonstrated in research about Generation Z's sustainability, in which attitudinal change through content leads to changes in purchase intention. [5].

> **H2:***Sustainability content exposure on short-form video platforms is positively associated with environmental attitude among Generation Z consumers.*

Beyond changing individual attitudes, sustainability content also shapes what people think their peers believe and expect. Research found that social media information sharing significantly raises subjective norms toward green consumption (β = 0.637; $p < 0.001$) [4], and short-form video strengthens this further by making sustainable behavior look normal and common within a user's peer group [5]. This kind of normative pressure is a separate internal response from awareness or attitude.

> **H3:***Sustainability content exposure on short-form video platforms is positively associated with perceived social norms toward sustainability among Generation Z consumers.*

Awareness does not just stay as background knowledge; it actually converts into purchase intention. Green consumer knowledge has been observed to have a direct positive influence on the purchase behavior of Generation Z individuals (β = 0.163) [1], and this finding has also been confirmed in a study involving Indonesian Generation Z(β = 0.122; $p < 0.001$) [9], which suggests this awareness-to-intention pathway works across different cultural contexts.

> **H4:***Environmental awareness is positively associated with sustainable purchase intention among Generation Z consumers.*

Prior studies identify attitude as a key predictor of purchase intention. Sustainability content attitude was found to be the strongest factor impacting the Generation Z purchase intention (β = 0.39; R^2 = 48%) [12]. A separate Generation Z study found a similar direct path from attitude to green behavior (β = 0.244;$p < 0.001$) [9], which supports attitude as a reliable pathway to purchase intention.

> **H5:***Environmental attitude is positively associated with sustainable purchase intention among Generation Z consumers.*

Peer norms work as an independent driver of purchasing behavior rather than a contextual variable. What peers expect matters regardless of what the individual personally knows or believes. Research confirms that normative pressure directly predicts the green purchasing intention (β = 0.552; $p < 0.001$), independently of personal attitude or awareness levels [4], which shows that peer expectations form their own distinct pathway to sustainable purchasing.

> **H6:***Perceived social norms toward sustainability are positively associated with sustainable purchase intention among Generation Z consumers.*

A. Conceptual Framework

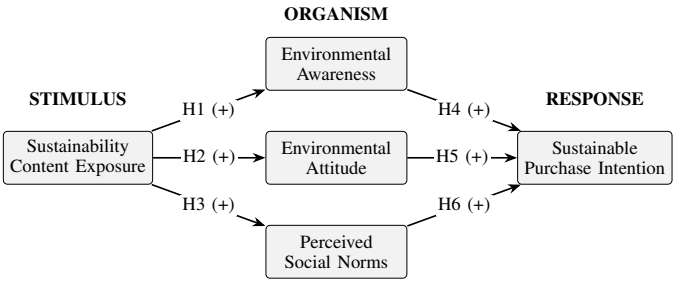


Fig. 1. Conceptual framework based on S-O-R theory.

Figure 1 shows the conceptual model that this study is built around. Sustainability content exposure acts as the stimulus in the SOR framework. This stimulus activates three organism-level variables: environmental awareness, environmental attitude, and perceived social norms, which together are expected to influence a person's intention to purchase sustainable products as the behavioral response [4], [5], [9]. The six hypotheses that emerge from this model are discussed in Section III.

IV. Methodology

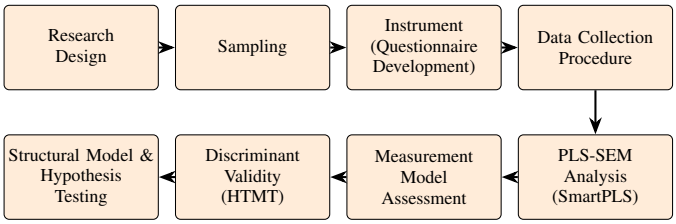


Fig. 2. Methodological workflow of the study.

A. Research Design

Figure 2 summarises the research workflow, from research design and questionnaire development through data collection, PLS-SEM analysis, measurement validation and hypothesis testing. This study adopts a quantitative cross-sectional survey design. Given that the hypotheses developed in Section III specify directional relationships between sustainability content exposure, three organism-level mediators, and sustainable purchase intention, a quantitative approach is required to statistically test the strength and significance of each proposed path. A qualitative approach would not permit the calculation of path coefficients (β) or significance levels (p-values) required for assessment of the conceptual framework presented in Section III-A.

B. Sampling

The population under consideration is made up of Generation Z Students studying at GISMA University, BSBI, Arden University, and the University of Europe(UE) in Germany. Such a choice of the population was made since international Generation Z students are frequent users of short-form video platforms and directly interact with sustainability content in different cultural environments, making them a suitable sample for examining how sustainability content influences purchase intention across different cultural contexts. Convenience sampling was chosen owing to the accessibility of the selected population and time limitations. The desired sample size is 100 respondents because this meets the minimum sample size recommended for PLS-SEM to produce stable and reliable path coefficient estimates [18], [19]. Additionally, the sample size of 100 fulfills the minimum R^2 criterion to estimate PLS-SEM sample size adequacy [18], which is based on Cohen's table of statistical power and takes into account the largest number of structural paths leading to one endogenous construct, the desired level of significance, and the minimum R^2 to detect; For a model with at most three precitors of one construct, a power analysis ($\alpha = 0.05$, power = 0.80) indicates a minimum of 77 observations to detect a medium effect ($f^2 = 0.15$) and 114 for a smaller effect ($f^2 = 0.10$). The sample of 100 is therefore adequate for medium effects but may be underpowered for small ones.

C. Instrument and Questionnaire Development

A self-administered questionnaire was created on Google Forms. It had six sections: (i) demographic information (age, gender, education, university and the platform on which the respondent most often sees sustainability content); (ii) sustainability content exposure (SCE); (iii) environmental awareness (EA); (iv) environmental attitude (AT); (v) perceived social norms (SN); and (vi) sustainable purchase intention (PI). Each of the six constructs was measured with five reflective items on a five-point agreement scale (1 = strongly disagree, 5 = strongly agree), giving 25 items in total (see the Appendix). The items were adapted from scales used in earlier studies: SCE from the content-exposure constructs in [5], [8], EA from the green consumer knowledge scale in [1], AT from the attitude construct in [9], SN from the subjective norms scale in [4], and PI from [9], [12]. The original wording was adapted to the study context. The exposure items refer explicitly to short-form video platforms (TikTok, Instagram Reels and YouTube Shorts), while the awareness items and most social norm and purchase intention items refer to social media or online content more broadly. The adapted items were checked for clarity and relevance by the authors before distribution. At the start of data collection, fewer than then responses collected through a university WhatsApp group were reviewed for clarity and completion. No changes to the instrument were needed, so these responses were kept in the final sample of 100.

D. Data Collection Procedure

The questionnaire was distributed by using university WhatsApp groups, Instagram, and peer networks of international students from the four universities that participated in the research. Participation in the research was completely voluntary, and the participants were notified about the aim of the research, its anonymity, and the possibility to terminate participation at any moment before submission according to the usual ethics of scientific research. No personally identifiable data was gathered in the process. Due to the fact that participants were chosen from university WhatsApp groups, Instagram, and referrals by other individuals, there was a potential for self-selection and network bias to occur because those individuals who decided to take part in the research might have differed significantly from those who did not respond due to their prior interest in sustainability topics or social media usage in general. This factor makes the findings less generalizable, and it is further analyzed as a limitation in Section VI-A.

E. Data Analysis Technique

The survey data analysis was done using PLS-SEM in the SmartPLS software package. The choice of this method over covariance-based structural equation modelling (CB-SEM) was made due to three special reasons connected with the study design. Firstly, the structure of the conceptual model includes three parallel mediators being simultaneously predicted by a single exogenous variable and predicting simultaneously a single endogenous variable, a structure better estimated by PLS-SEM than CB-SEM. Secondly, the size of the obtained sample(N=100), although being sufficient for PLS-SEM analysis according to usual criteria, is too small for asymptotic procedures used in CB-SEM. Finally, this research

aims to explain variance in the dependent variable (R^2) and compare the strengths of different predictor paths, which corresponds better to the estimation approach of PLS-SEM than the confirmatory theory testing approach of CB-SEM. The data analysis was done in two steps. Firstly, the evaluation of the outer model(reliability and validity: Cronbach's alpha, composite reliability, average variance extracted). Secondly, the evaluation of the inner model (path coefficient (β), significance levels ($p < 0.05$), and the coefficient of determination (R^2) for the endogenous variables.

F. Model Specification and Evaluation Metrics

Each construct is modeled as reflective, so every observed item is a linear function of its latent construct plus measurement error:

$$x_{jk} = \lambda_{jk}\,\xi_j + \varepsilon_{jk}, \tag{1}$$

where x_{jk} is item k of construct j, ξ_j is the latent construct, λ_{jk} is the outer loading and ε_{jk} is the measurement error. The structural model follows the S-O-R chain in Fig. 1:

$$\begin{aligned} \mathrm{EA} &= \gamma_1\,\mathrm{SCE} + \zeta_1, \\ \mathrm{AT} &= \gamma_2\,\mathrm{SCE} + \zeta_2, \\ \mathrm{SN} &= \gamma_3\,\mathrm{SCE} + \zeta_3, \\ \mathrm{PI} &= \beta_1\,\mathrm{EA} + \beta_2\,\mathrm{AT} + \beta_3\,\mathrm{SN} + \zeta_4, \end{aligned} \tag{2}$$

where γ_1 to γ_3 are the paths tested in H1 to H3, β_1 to β_3 are the paths tested in H4 to H6, and ζ denotes the residual of each equation. Because the model has no direct path from exposure to purchase intention, the specific indirect effect through mediator j is the product of the two paths on that route:

$$\mathrm{SCE} \to j \to \mathrm{PI}: \quad \gamma_j\,\beta_j, \qquad j = 1, 2, 3. \tag{3}$$

Reliability and convergent validity are assessed with composite reliability and average variance extracted (AVE):

$$\rho_c = \frac{\left(\sum_k \lambda_k\right)^2}{\left(\sum_k \lambda_k\right)^2 + \sum_k \left(1 - \lambda_k^2\right)}, \tag{4}$$

$$\mathrm{AVE} = \frac{1}{K}\sum_{k=1}^{K} \lambda_k^2, \tag{5}$$

where K is the number of items of the construct. Values above 0.70 for ρ_c and above 0.50 for AVE are considered acceptable [18]. Discriminant validity is assessed with the heterotrait-monotrait ratio:

$$\mathrm{HTMT}_{jl} = \frac{\bar{r}_{jl}}{\sqrt{\bar{r}_j\,\bar{r}_l}}, \tag{6}$$

where $\bar{r}_{jl}$ is the mean correlation between the items of construct j and l, and $\bar{r}_j$ and $\bar{r}_l$ are the mean correlations between the items in the respective constructs. Values below 0.90 indicate discriminant validity [20]. The effect size of a predictor and multicollinearity among predictors are computed as

$$f^2 = \frac{R^2_{\mathrm{incl}} - R^2_{\mathrm{excl}}}{1 - R^2_{\mathrm{incl}}}, \tag{7}$$

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}, \tag{8}$$

where R^2_{incl} and R^2_{excl} are the R^2 of the endogenous construct with and without the predictor, and R_j^2 comes from regressing predictor j on the other predictors. Eqs. (4) and (5) are reported in Table II, Eq. (6) in Table IV, Eqs. (2) and (7) in Table V, Eq. (3) in Table VI, and Eq. (8) in Section V-E.

V. Results

A. Sample Characteristics

A total of 100 Generation Z participants were involved in this research. The biggest age segment was 24-27 years (49%), followed by 21-23 years old (35%), while the relatively lower segment includes participants aged 18-20 (9%) and 28 or above (7%). Women participants constituted the majority in the sample (50%), with males (36%), others (13%), and 1% participants who prefer not to state their gender. When considering education, 79% respondents have postgraduate qualifications, and 20% were undergraduates, which is a reflection of the academic nature of the sample target population. As per platform usage, Instagram (51%) was the primary platform used, followed by YouTube Shorts (30%), TikTok (17%), and other platforms (2%). Complete sample characteristic details have been provided in Table I.

TABLE I
SAMPLE CHARACTERISTICS (N = 100)

Variable	Category	Frequency	Percentage
Age	18–20	9	9%
	21–23	35	35%
	24–27	49	49%
	28+	7	7%
Gender	Female	50	50%
	Male	36	36%
	Non-binary/other	13	13%
	Prefer not to say	1	1%
Education	Postgraduate	79	79%
	Undergraduate	20	20%
	Other	1	1%
Platform	Instagram	51	51%
	YouTube Shorts	30	30%
	TikTok	17	17%
	Other	2	2%

Note. Platform = the platform on which the respondent most often encounters sustainability content.

B. Measurement Model Assessment

Prior to examining structural relationship analysis, the reliability and validity of all constructs have been considered. As evident from Table II, the Cronbach's alpha values ranged from 0.848 to 0.888, and comfortably exceeded the threshold of 0.70, signifying a high level of consistency in all five constructs. Composite reliability value (ρ_c) ranged from 0.892 to 0.917, thereby reaffirming the fact that each construct reliably measures what it is supposed to measure. Average variance extracted (AVE) values ranged from 0.623 to 0.690, and all

were above the 0.50 benchmark, signifying that constructs account for a much greater proportion of variance in indicators as compared to measurement error [18], [21]. Overall, the results suggested that the measurement model was reliable and had satisfactory convergent validity. Table III presents the

TABLE II
MEASUREMENT MODEL ASSESSMENT

Construct	Cronbach's α	ρ_c	AVE
AT	0.848	0.892	0.623
EA	0.853	0.895	0.631
PI	0.888	0.917	0.690
SCE	0.849	0.892	0.625
SN	0.880	0.913	0.677

Note. Thresholds: $\alpha > 0.70$, $\rho_c > 0.70$, AVE > 0.50 [18].

individual item loadings for each construct. All items loaded above the recommended 0.7 threshold [18], indicating strong indicator reliability across all five constructs. Item means ranged from 3.70 to 4.15 and item standard deviations from 1.05 to 1.41 on the five-point scale.

C. Discriminant Validity

To examine whether the constructs were sufficiently distinct from one another, the HTMT criterion was applied with a 95% bootstrapped confidence interval. A value below 0.90 was used as the cut off for acceptable discriminant validity [18]. As shown in Table IV, six of ten construct pairs fall within the acceptable range. Four pairs exceed the threshold: EA $\leftrightarrow$ AT (1.052), SCE $\leftrightarrow$ EA (1.056), SCE $\leftrightarrow$ AT (1.007), and SN $\leftrightarrow$ PI (1.010). The correlation between EA and AT is consistent with the strong conceptual relationship between these two factors and their correlation with each other at the measurement stage in sustainability studies [9]. The correlation of exposure with EA and AT and with purchase intention indicates that participants did not understand the difference between these concepts; this is considered a limitation in Section VI-A. Table IV shows the 95% bias-corrected bootstrap confidence intervals for all pairs of constructs. For five construct pairs (EA-AT, SCE-EA, SCE-AT, SN-PI and AT-PI), the confidence interval contains 1, thus making them indistinguishable from each other. In the case of the other five pairs, the confidence upper limit was less than 1 but more than 0.90. The correlation coefficient for the constructs SCE and EA (0.901) is greater than the square root of the average variance extracted (AVE) for both the constructs (0.790 and 0.794), which validates the overlap. Thus, discriminant validity is not demonstrated completely, and conclusions drawn from these pairs have to be taken cautiously (Section VI-A).

As a sensitivity analysis, EA and AT, the pair with the highest HTMT value, were aggregated into a single construct of ten items, and HTMT was calculated again. Consequently, the number of pairs of constructs with the value above 0.90 was reduced from four to two, as sustainability content exposure was still highly associated with the new construct (HTMT $= 1.017$), and the correlation between social norms and purchase intention did not change (HTMT $= 1.010$). However, the construct was internally consistent (Cronbach's $\alpha = 0.924$, all item-total correlations > 0.60). Thus, the overlap between EA and AT is a result of construct redundancy, whereas the overlap between exposure and internal states and between social norms and purchase intention persists despite the item aggregation and should be considered a structural issue.

TABLE III
OUTER LOADINGS AND ITEM STATISTICS (LOADING; MEAN (SD), N = 100)

Item	AT	EA	PI	SCE	SN
	0.758	0.816	0.807	0.805	0.874
1	4.14 (1.22)	4.15 (1.18)	3.87 (1.28)	4.06 (1.38)	3.81 (1.41)
	0.773	0.771	0.841	0.785	0.790
2	4.12 (1.16)	3.82 (1.19)	3.90 (1.28)	3.96 (1.06)	3.70 (1.34)
	0.832	0.820	0.819	0.850	0.857
3	3.84 (1.37)	4.06 (1.14)	3.80 (1.20)	3.86 (1.18)	3.74 (1.30)
	0.764	0.731	0.824	0.727	0.781
4	4.08 (1.13)	3.95 (1.09)	3.83 (1.24)	4.04 (1.05)	3.78 (1.27)
	0.816	0.829	0.861	0.779	0.807
5	4.10 (1.22)	4.01 (1.23)	3.89 (1.27)	4.08 (1.35)	3.75 (1.31)

Note. Each cell shows the outer loading (top) and the item mean with SD in parentheses (bottom). Item k belongs to the construct in its column. Scale: 1 = strongly disagree, 5 = strongly agree. All loadings exceed 0.70 [18].

TABLE IV
HTMT RATIOS WITH 95% BOOTSTRAP CONFIDENCE INTERVALS

Pair	HTMT	95% CI	CI includes 1
EA $\leftrightarrow$ AT	1.052*	[0.971, 1.150]	Yes
SCE $\leftrightarrow$ EA	1.056*	[0.982, 1.152]	Yes
SCE $\leftrightarrow$ AT	1.007*	[0.866, 1.128]	Yes
SN $\leftrightarrow$ PI	1.010*	[0.944, 1.077]	Yes
PI $\leftrightarrow$ AT	0.855	[0.672, 1.007]	Yes
SCE $\leftrightarrow$ PI	0.799	[0.582, 0.972]	No
SN $\leftrightarrow$ SCE	0.831	[0.647, 0.969]	No
SN $\leftrightarrow$ AT	0.809	[0.622, 0.952]	No
SN $\leftrightarrow$ EA	0.797	[0.577, 0.947]	No
PI $\leftrightarrow$ EA	0.772	[0.539, 0.959]	No

Note. * HTMT > 0.90. 95% bias-corrected and accelerated bootstrap intervals (5,000 subsamples). AT = environmental attitude; EA = environmental awareness; PI = purchase intention; SCE = sustainability content exposure; SN = perceived social norms.

D. Structural Model and Hypotheses Testing

A structural model assessment was conducted through PLS-SEM bootstrapping analysis using 5,000 subsamples. The final structural model is shown in Figure 3 and the results are reported in Table V, both are consistent with the hypotheses of the research. Five out of six paths produced significant results. The significant path coefficients for the effects SCE has on all internal variables were: awareness ($\beta = 0.901$), attitude ($\beta = 0.854$), and social norms ($\beta = 0.719$). As for the variance explained by the model, awareness accounted for 81.1%, attitude for 72.9%, and social norms for 51.7%, while purchase intention had the highest R^2 at 84.1%. Together, the mediators explain a large share of the variance in purchase intention in this sample, although this high R^2 should be interpreted with caution given the collinearity reported in Section V-E. Predictive relevance was assessed with 10-fold cross-validation repeated 10 times. Q^2_{predict} was positive for all 20

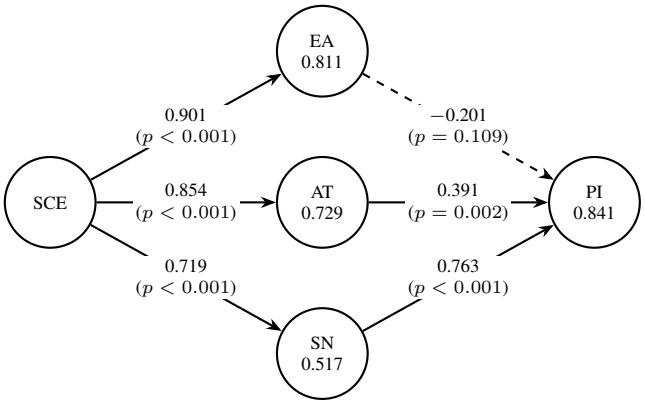


Fig. 3. PLS-SEM final structural model. Values on the arrows are standardized path coefficients (β) with p-values in parentheses; values inside the circles are R^2. The dashed arrow is not significant. Results are based on bootstrapping with 5,000 subsamples. SCE = sustainability content exposure; EA = environmental awareness; AT = environmental attitude; SN = perceived social norms; PI = purchase intention.

TABLE V
STRUCTURAL MODEL RESULTS, EFFECT SIZES AND 95% CONFIDENCE INTERVALS

Hyp.	Path	β	t	p	95% CI	f^2	Decision
H1	SCE → EA	0.901	29.201	<0.001	[0.828, 0.948]	4.289	Supported
H2	SCE → AT	0.854	16.077	<0.001	[0.729, 0.935]	2.690	Supported
H3	SCE → SN	0.719	9.817	<0.001	[0.559, 0.845]	1.072	Supported
H4	EA → PI	−0.201	1.602	0.109	[−0.407, 0.087]	0.048	Not supported
H5	AT → PI	0.391	3.091	0.002	[0.117, 0.601]	0.177	Supported
H6	SN → PI	0.763	8.182	<0.001	[0.527, 0.892]	1.792	Supported

Note. β = standardized path coefficient. CIs are 95% bias-corrected bootstrap intervals (5,000 subsamples). $f^2 \geq 0.02$ small, ≥ 0.15 medium, ≥ 0.35 large [18].

indicators of the endogenous constructs (EA: 0.44 to 0.56; AT: 0.40 to 0.47; SN: 0.31 to 0.38; PI: 0.20 to 0.42). The PLS-SEM prediction error was lower than the linear model benchmark for 19 of the 20 indicators, which indicates medium predictive power [22]. One path was not significant: Hypothesis H4 was not supported because environmental awareness was not significantly related to purchase intention ($\beta = -0.201$, $p = 0.109$). This is consistent with the attitude-behaviour gap literature, which suggests that awareness alone is not enough to drive behavioural intentions. Since this model assumes that the three mediators are parallel in the mediation process and does not have a direct path between EA and AT, this study cannot examine whether awareness influences intentions via attitude. Moreover, the negative EA → PI path can also be interpreted cautiously since there is high collinearity between EA and AT (VIF > 5, Section V-E) and their HTMT ratio is higher than 1.

Table V also report the effect size (f^2) of each path. Sustainability content exposure had a very large effect on all three mediators (all $f^2 > 1$), reflecting the strong path coefficient above. Among the predictors of purchase intention, perceived social norms had the largest effect ($f^2 = 1.792$), followed by environmental attitude with a medium effect ($f^2 = 0.177$) and environmental awareness with a small effect ($f^2 = 0.048$).

As awareness, attitude, and social norms are mediators paralleling each other between sustainability content exposure and purchase intention according to the proposed model, the indirect effects were assessed in detail (Table VI). The indirect effect via social norms ($\beta = 0.549$, $p < 0.001$) and attitude ($\beta = 0.334$, $p = 0.001$) was found to be significant, but the indirect effect via awareness was not ($\beta = -0.181$,$p = 0.112$, 95% CI [−0.397, 0.040]). The total indirect effect of sustainability content exposure on purchase intention was found to be significant($\beta = 0.702$, $p < 0.001$). As there is no direct link from exposure to purchase intention in the proposed model, this means that exposure influences purchase intention via social norms and attitude but not via awareness.

TABLE VI
SPECIFIC INDIRECT EFFECTS OF SCE ON PI

Path	β	t	p	95% CI
SCE → SN → PI	0.549	8.803	<0.001	[0.432, 0.667]
SCE → AT → PI	0.334	3.180	0.001	[0.109, 0.519]
SCE → EA → PI	−0.181	1.590	0.112	[−0.397, 0.040]
Total indirect effect	0.702	8.645	<0.001	[0.519, 0.837]

Note. 95% bias-corrected and accelerated bootstrap intervals (5,000 subsamples).

E. Collinearity and Common Method Bias

Collinearity among the predictors was found using the inner-model VIF approach. The VIF for awareness and attitude when explaining purchase intention was 5.313 (EA) and 5.454 (AT), while for social norms, the VIF was 2.046. The three paths from sustainability content exposure contained only one predictor; the VIF for them is 1.000. For EA and AT, VIFs exceeded the cutoff value of 5.0 for PLS-SEM [18]. As all the constructs were measured through one self-reporting questionnaire, common method variance was checked in two ways. First, according to Harman's single factor test, the percentage of variance accounted for by a single factor was found to be 53.2%, which exceeds the 50% cutoff. Further, the full collinearity VIFs for the five constructs ranged from 5.92 to 7.97, exceeding the 3.3 cutoff value recommended by Kock [23]. Common method variance therefore cannot be ruled out, and it may have influenced the strength of the path coefficients and the overlap between constructs.

F. Platform Comparison

To examine whether the levels of the constructs differ by platform, the constructed scores(means of item scores on the five-point scale) were compared across the three main platform categories as reported in Table VII. Subjects who selected the category "Other" ($n = 2$) were excluded. Using the Kruskal-Wallis test, there were no significant differences among Instagram, YouTube Shorts, and TikTok users for any of the constructs (all $p > 0.05$). Since the TikTok group size is small ($n = 17$), such tests are exploratory in nature. To explore whether the relationship between exposure and purchase intention differs by platform, Spearman correlations

were computed within each platform group. Sustainability content exposure was positively correlated with purchase intention on all platforms (Instagram: $r_s = 0.58$, $n = 51$; YouTube Shorts: $r_s = 0.90$, $n = 30$; TikTok: $r_s = 0.80$, $n = 17$; all $p < 0.001$), although the association appeared weaker among Instagram users. Subgroup sizes were too small for formal multi-group analysis, so these results are exploratory.

TABLE VII
CONSTRUCT SCORES BY PLATFORM (MEAN ± SD) AND KRUSKAL-WALLIS TEST

	Instagram (n=51)	YouTube Shorts (n=30)	TikTok (n=17)	H	p
SCE	4.15 ± 0.82	3.76 ± 1.07	4.20 ± 0.81	3.80	0.150
EA	4.11 ± 0.82	3.80 ± 1.06	4.22 ± 0.62	2.20	0.332
AT	4.22 ± 0.79	3.83 ± 1.16	4.21 ± 0.66	1.98	0.371
SN	3.78 ± 1.12	3.69 ± 1.11	3.98 ± 0.82	1.02	0.601
PI	3.91 ± 1.04	3.81 ± 1.10	4.02 ± 0.74	0.11	0.945

Note. Construct scores are means of item scores on the five-point scale. H and p are from Kruskal-Wallis tests (2 degrees of freedom). Respondents who chose "Other" ($n = 2$) were excluded.

VI. DISCUSSION

Exposure to sustainability content on short-form video was positively associated with environmental awareness (H1: $\beta = 0.901$), environmental attitude (H2: $\beta = 0.854$) and perceived social norms (H3: $\beta = 0.719$), all $p < 0.001$. These coefficients are larger than those reported for social media exposure in earlier work, for example $\beta = 0.281$ for exposure and sustainability intention [8] and $\beta = 0.637$ for information sharing and subjective norms [4]. Part of the difference may reflect construct overlap: HTMT between exposure and both awareness and attitude exceeded 1, so these constructs are not clearly distinct in this sample (Section V-C), and the three paths are best read as strong associations and not as separate effects. Among the paths to purchase intention, perceived social norms was the strongest (H6: $\beta = 0.763$, $f^2 = 1.792$), followed by environmental attitude (H5: $\beta = 0.391$, $p = 0.002$). The attitude coefficient is almost identical to the one found for TikTok users in the United States ($\beta = 0.39$) [12] and higher than in Indonesian Generation Z ($\beta = 0.244$) [9]. The norm coefficient is higher than the $\beta = 0.552$ reported by [4]. Awareness was not significant (H4: $\beta = -0.201$, $p = 0.109$), unlike the small positive effects reported earlier [1], [9]. This agrees with the attitude-behaviour gap, and the negative sign likely reflects overlap with attitude (HTMT $= 1.052$, VIF > 5). Exposure was related to purchase intention through social norms ($\beta = 0.549$) and attitude ($\beta = 0.334$) but not through awareness ($\beta = -0.181$, $p = 0.112$). Social norms, rather than attitude, were the strongest predictor, unlike earlier work [9], [12].

A possible reason is the peer-oriented situation of international students, although this was not tested. Because the HTMT between social norms and purchase intention was 1.010, this result also needs to be confirmed with items that separate peer expectations from personal intention more clearly. Platform results are exploratory. Construct scores did not differ significantly between Instagram, YouTube Shorts, and TikTok users (all $p > 0.05$), and exposure was positively correlated with purchase intention on all three platforms ($r_s = 0.58$ to 0.90). With only 17 TikTok respondents and a platform variable that records where respondents most often see sustainability content, the data cannot show that one platform is more effective than another. In practical terms, brands and communicators who target Generation Z may get more from content that shows peers choosing sustainable products than from content that only informs. Since the design is cross-sectional, these are associations and not causal effects. The findings are relevant to SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action), as they point to peer-based short-form content as one way to encourage responsible consumption among young consumers.

A. Limitations

This study has a number of limitations that must be considered in conjunction with the results obtained. The convenience sample was recruited through university WhatsApp groups, Instagram and peer referrals, so self-selection bias is likely, and the sample was dominated by postgraduate students (79%) at four German universities, which means the results may not represent Generation Z consumers of diverse backgrounds. With $N = 100$, the model is adequate for medium effects but may miss small ones. Instagram was the primary platform for over half of the sample (51%), and only 17 respondents used TikTok, which means the findings may reflect Instagram user behaviour more than TikTok or YouTube Shorts users and that the platform groups were too small for formal comparison. Four pairs of constructs had HTMT values above 0.90 (Section V-C), so discriminant validity is not fully established, and awareness and attitude were highly collinear (VIF > 5). All data come from one self-reported questionnaire, and Harman's test (53.2%) means common method variance cannot be ruled out. The survey took place at one point in time and measured intention, not actual purchase, so it cannot show causation or the gap between intention and buying behaviour. Future research should use larger cross-national samples and behavioural data.

VII. CONCLUSION

The present paper explored how exposure to sustainability content on short-form video platforms (TikTok, Instagram Reels and YouTube Shorts) relates to the purchase intention of Generation Z consumers, using the S-O-R model. Five of the six hypotheses were supported. Exposure was strongly associated with environmental awareness, environmental attitude, and perceived social norms. Among the three, perceived social norms was the strongest predictor of purchase intention, followed by environmental attitude, while environmental awareness had no significant effect. Platform comparisons showed no significant differences in construct scores, and exposure was positively related to purchase intention on all three platforms. Whether the platforms differ in effectiveness remains an open question for future research. These findings

are relevant to SDG 12 and SDG 13. They should be read in light of the study's limitations, in particular the overlap between several constructs and the convenience sample. Future research could use a larger cross-national sample, measure platform-specific exposure and actual purchases, and refine the items so that awareness, attitude, and exposure are empirically distinct.

APPENDIX

TABLE VIII
MEASUREMENT ITEMS (FIVE-POINT SCALE, 1 = STRONGLY DISAGREE, 5 = STRONGLY AGREE)

Code	Item wording
SCE1	I regularly come across sustainability-related content (e.g. eco-friendly products, green living, climate awareness) while using short-form video platforms.
SCE2	Short-form video platforms (TikTok, Instagram Reels, or YouTube Shorts) frequently show me content related to environmental issues or sustainable consumption.
SCE3	The sustainability content I see on short-form videos feels relevant to my daily life.
SCE4	The short-form video platform I use most often exposes me to sustainability content through its algorithm.
SCE5	Over the past month, I have encountered a significant amount of sustainability-related content on short-form video platforms.
EA1	I can explain the environmental impact of common consumer products because of content I have seen online.
EA2	Because of social media content, I know which everyday habits contribute most to environmental damage.
EA3	I can identify which products are more environmentally friendly as a result of content I have seen on social media.
EA4	Social media has helped me understand the consequences of unsustainable consumption.
EA5	I am more informed about environmental problems because of sustainability content I follow online.
AT1	I feel guilty when I buy products that harm the environment.
AT2	I feel proud when I make environmentally responsible purchasing decisions.
AT3	I feel uncomfortable knowing my purchases may damage the environment.
AT4	I feel positive emotions when I choose eco-friendly products over conventional ones.
AT5	I feel that living sustainably is personally important to me.
SN1	People around me expect me to make environmentally responsible purchasing decisions.
SN2	My peers who follow sustainability content on social media influence my purchasing choices.
SN3	I feel social pressure from my online community to choose sustainable products.
SN4	Sustainability trends on short-form video platforms make sustainable purchasing feel like the norm among my peers.
SN5	I am more likely to buy sustainable products when I see others doing the same on social media.
PI1	I intend to purchase environmentally friendly products in the near future.
PI2	Sustainability content I have seen on social media has increased my intention to buy green products.
PI3	I am willing to pay a premium for products that are environmentally sustainable.
PI4	I would choose a sustainable product over a conventional one if both were available.
PI5	I plan to actively seek out sustainable product options when making my next purchase.

REFERENCES

[1] P. S. Borah, C. S. K. Dogbe, and N. Marwa, "Generation Z's green purchase behavior: Do green consumer knowledge, consumer social responsibility, green advertising, and green consumer trust matter for sustainable development?" *Business Strategy and the Environment*, vol. 33, no. 5, pp. 4530–4546, 2024.

[2] K. Vladimirova, C. E. Henninger, S. I. Alosaimi, T. Brydges, H. Choopani, M. Hanlon, S. Iran, H. McCormick, and S. Zhou, "Exploring the influence of social media on sustainable fashion consumption: A systematic literature review and future research agenda," *Journal of Global Fashion Marketing*, vol. 15, no. 2, pp. 181–202, 2024.

[3] I. Mirza, Z. Javed, and R. K. Malhi, "The impact of digital apps in shaping sustainable consumption behavior: A study on Gen Z from Bhopal," in *Proceedings of the 5th International Conference on the Role of Innovation, Entrepreneurship and Management for Sustainable Development (ICRIEMSD 2024)*, vol. 310. Springer Nature, 2024, pp. 425–442.

[4] S. Xie and G. R. Madni, "Impact of social media on young generation's green consumption behavior through subjective norms and perceived green value," *Sustainability*, vol. 15, no. 4, p. 3739, 2023.

[5] M. G. Confetto, C. Covucci, F. Addeo, and M. Normando, "Sustainability advocacy antecedents: How social media content influences sustainable behaviours among Generation Z," *Journal of Consumer Marketing*, vol. 40, no. 6, pp. 758–774, 2023.

[6] A. Mehrabian and J. A. Russell, *An Approach to Environmental Psychology*. Cambridge, MA: MIT Press, 1974.

[7] A. Horrich, M. Ertz, and I. Bekir, "The effect of information adoption via social media on sustainable consumption intentions: The moderating influence of gender," *Current Psychology*, vol. 43, no. 18, pp. 16 349–16 362, 2024.

[8] T. Hidayat, A. F. Nashirah, and I. R. D. Pangestuti, "Analyzing the consumer behavior and implications of sustainable consumption among Generation Z," *Research Horizon*, vol. 4, no. 4, pp. 161–176, 2024.

[9] E. F. Adah, A. Nugroho, W. Rachbini *et al.*, "Closing the green gap: Attitude drives green consumer behavior among Gen Z in West Java," *TEC EMPRESARIAL*, vol. 20, no. 2, pp. 866–896, 2025.

[10] A. Shaheen Hosany and K. Serdiuk, "Understanding Gen Z consumers: A typology of (Un)sustainable purchases," *Psychology & Marketing*, vol. 42, no. 11, pp. 2820–2832, 2025.

[11] A. Halibas, U. Akram, A.-P. Hoang, and M. D. Thi Hoang, "Unveiling the future of responsible, sustainable, and ethical consumption: A bibliometric study on Gen Z and young consumers," *Young Consumers*, vol. 26, no. 7, pp. 142–171, 2025.

[12] C. A. Lin, X. Wang, and L. Dam, "TikTok videos and sustainable apparel behavior: Social consciousness, prior consumption and theory of planned behavior," *Emerging Media*, vol. 1, no. 1, pp. 46–69, 2023.

[13] S. Plötz, L. M. Martinez, L. F. Martinez, and F. R. Ramos, "The influence of TikTok videos on German Gen Z consumers' attitude and purchase intention towards sustainable brands," in *Digital Marketing & eCommerce Conference*. Springer, 2023, pp. 270–289.

[14] D. El-Shihy and S. Awaad, "Leveraging social media for sustainable fashion: How brand and user-generated content influence Gen Z's purchase intentions," *Future Business Journal*, vol. 11, p. 113, 2025.

[15] M. Pittman and Q. Li, "Green advertising on Instagram versus TikTok: Construal level moderates processing fluency and environmental commitment," *Journal of Interactive Advertising*, 2025, in press.

[16] M. I. Seeliga, A. Shata, Z. Yang, D. Sd, Y. Gao, H. Hu, and J. Yang, ""making the environment like a cool thing:" exploring Generation Z and Millennials' reception of climate change videos on YouTube," *Atlantic Journal of Communication*, vol. 34, no. 2, pp. 181–199, 2025.

[17] I. Ajzen, "The theory of planned behavior," *Organizational Behavior and Human Decision Processes*, vol. 50, no. 2, pp. 179–211, 1991.

[18] J. F. Hair, J. J. Risher, M. Sarstedt, and C. M. Ringle, "When to use and how to report the results of PLS-SEM," *European Business Review*, vol. 31, no. 1, pp. 2–24, 2019.

[19] N. Kock and P. Hadaya, "Minimum sample size estimation in PLS-SEM: The inverse square root and gamma-exponential methods," *Information Systems Journal*, vol. 28, no. 1, pp. 227–261, 2018.

[20] J. Henseler, C. M. Ringle, and M. Sarstedt, "A new criterion for assessing discriminant validity in variance-based structural equation modeling," *Journal of the Academy of Marketing Science*, vol. 43, no. 1, pp. 115–135, 2015.

[21] C. Fornell and D. F. Larcker, "Evaluating structural equation models with unobservable variables and measurement error," *Journal of Marketing Research*, vol. 18, no. 1, pp. 39–50, 1981.
[22] G. Shmueli, M. Sarstedt, J. F. Hair, J.-H. Cheah, H. Ting, S. Vaithilingam, and C. M. Ringle, "Predictive model assessment in PLS-SEM: Guidelines for using PLSpredict," *European Journal of Marketing*, vol. 53, no. 11, pp. 2322–2347, 2019.
[23] N. Kock, "Common method bias in PLS-SEM: A full collinearity assessment approach," *International Journal of e-Collaboration*, vol. 11, no. 4, pp. 1–10, 2015.